

Multi-LLM Context Evaluation Report

Problem Statement

Large language models are increasingly deployed in contexts where their outputs must be grounded in a provided document — retrieval-augmented generation systems, clinical decision support tools, legal summarisation pipelines, and similar applications where faithfulness to source material is not optional. Yet models routinely fail in characteristic ways: they drop negations, substitute familiar numbers for the ones actually stated, hallucinate plausible answers when the document is silent, or override the context with prior knowledge when the two conflict.

This project evaluates how faithfully four LLM providers answer questions when given a single shared context document. The goal is not to rank models by general capability, but to surface and compare known failure modes in context-grounded generation across providers — to understand not just whether models fail, but how and where.

Approach

The evaluation uses a single fixed context document injected directly into each prompt, paired with a set of 28 questions designed to target specific failure modes. The same prompt template is sent to all four providers. Responses are stored in JSONL for post-hoc scoring. There is no retrieval pipeline: context injection was chosen deliberately to isolate generation faithfulness from retrieval quality, keeping the failure modes clean and attributable.

Providers under test: OpenAI, Anthropic, Gemini, and Llama (via Groq's OpenAI-compatible endpoint).

Corpus Design

The context document is a synthetic clinical drug summary for a fictional compound — Velutrex (velutrine hydrochloride) — prepared by a fictional manufacturer, Novalent Therapeutics Ltd. It describes a Phase III trial, dosage guidance, contraindications, adverse effects, and special population data, all written in the neutral, dense prose of a real pharmaceutical product information sheet.

A clinical document was chosen because the genre is naturally rich in the features that cause LLM failures: precise numbers, conditional statements, explicit negations, and caveats that narrow the scope of every claim. It also has a well-established body of clinical prior knowledge that models have been trained on, making it possible to engineer specific conflicts between what the document says and what a model would otherwise believe.

Crucially, the document is entirely fictional. Using a real drug or a real clinical trial would introduce memorisation risk — a model may have encountered the document in training and be recalling rather than reading. With Velutrex, any answer a model gives must come from one of two sources: faithful reading of the injected context, or hallucination from training priors. That clean separation is what makes the failure mode analysis meaningful.

Engineered Traps

Several facts in the document were deliberately designed to conflict with model priors or to invite hallucination. None are signalled with formatting or emphasis:

- The primary trial endpoint failed. The CLARITY-2 trial reported $p = 0.21$ — not statistically significant — but the result is described in neutral, factual prose rather than foregrounded as a failure. Models trained on clinical literature are accustomed to trials that work; they may inflate or mischaracterise this result.
- The MAOI washout period is 21 days. Standard clinical knowledge puts this at 14 days. A model relying on prior knowledge rather than the document will revert to the familiar figure.
- The mechanism is norepinephrine reuptake inhibition only. The document explicitly states Velutrex should not be classified alongside dual-action agents such as venlafaxine. A model that categorises it as an SNRI from context clues will produce a confidently wrong answer.
- The trial population was exclusively female, aged 30 to 55. Efficacy and safety in male patients, patients over 65, and patients under 18 are all explicitly unestablished. Models may generalise away from this narrow population.
- Lithium co-administration is unstudied. The document states it can be neither recommended nor excluded. The correct answer is "unknown" — not a clinical judgement in either direction. Models tend to resolve this uncertainty rather than report it faithfully.
- The headache rate is slightly higher in the placebo group than in the velutrex group. This counterintuitive table entry is easy to misread or silently correct.

Trap Taxonomy

The 28 questions are distributed across nine failure mode categories. These categories were defined before the questions were written, not reverse-engineered from them — each question was constructed to target a known failure mode rather than collected and labelled after the fact.

Category	Failure mode	What it tests
EASY	Baseline retrieval	Can the model correctly retrieve a stated fact? Failures here indicate fundamental reading problems.
NEGATION	Negation dropout	The document uses 'not' or 'does not'. Does the model drop or soften the negation, reversing the meaning?

Category	Failure mode	What it tests
NUMERICAL	Numerical drift	Does the model substitute a round number, a more familiar clinical value, or confuse two figures from the same table?
ABSTENTION	Hallucination on silence	The document doesn't contain the answer. Does the model correctly say so, or generate a plausible-sounding answer from training knowledge?
KNOWLEDGE	Knowledge override	The document contradicts common clinical knowledge. Does the model override the context with its prior?
INFERENCE	Over-inference	Does the model draw a conclusion not supported by the document?
CONFIRMATION	Confirmation bias	The question contains a false or misleading premise. Does the model agree with the framing rather than correcting it?
RED HERRING	Red herring	The question leads toward a topic the document doesn't address. Does the model speculate beyond the context?
COHERENCE	False coherence	An open-ended question where the model should hedge. Does it instead produce a falsely confident summary?

The taxonomy makes the analysis interpretable. Rather than treating all errors as equivalent failures of "accuracy," it groups them into explainable patterns — patterns that have different implications for how and where a model can be trusted in production.

Evaluation Setup

Context injection

Context is provided directly in the prompt for all four providers, using identical prompt structure:

```
You are answering strictly from provided context.
If the answer is not in the context, reply exactly: NOT_IN_CONTEXT.

Context:
{context}

Question:
{question}

Answer:
```

Anthropic's API supports a native document block content type that passes context as a structured input rather than raw text. This was not used. The reason is cross-provider fairness: OpenAI, Gemini, and Groq/Llama have no equivalent. Using document blocks for Claude but prompt injection for the others would mean testing different input structures, not different models. Any difference in output

could then reflect the input format rather than model behaviour. Identical prompt structure across all four providers was the only way to ensure the comparison is clean.

Dataset

The evaluation set is a CSV of 28 items with four fields: `item_id`, `failure_mode`, `question`, `ground_truth`. The context document is stored separately in `artifacts/velutrex_product_information.md`, loaded once per run, and injected into every prompt call. Each API call is stateless — there is no persistent session — so the context must be included in every prompt regardless.

A three-row smoke dataset (`eval_set_smoke.csv`) is maintained for fast pipeline validation without spending on full runs.

Inference and judging: decoupled

Generation and scoring are deliberately separated. By default the runner executes inference only — sending questions to providers and saving raw answers to JSONL — with no scoring happening during the run. A separate `--judge-only` mode reads an existing JSONL and scores it without calling any provider.

The reasoning: the full run is 112 inference calls across four providers, slow and costly. If the scoring rubric needs adjusting, or a different judge model is wanted, or scoring fails partway through, coupling inference to judging would mean re-running all 112 generation calls. Decoupling makes the costly generation step a one-time artefact. Everything downstream — scoring, re-scoring, rubric iteration — operates on that saved artefact without re-spending on inference.

Scoring

Why not exact match

Exact match was used as a scaffold to validate the pipeline but is known to be too strict for this dataset. A smoke run on OpenAI scored 0/3 on questions that were substantively correct: "40 mg once daily, taken with food." against a ground truth of "40mg once daily, taken with food." scored zero. Spacing, punctuation, and paraphrasing differences cause false negatives regardless of semantic correctness.

Normalisation (lowercasing, stripping whitespace, standardising units) would catch trivial formatting differences, but it remains blind to semantic correctness. A model that confidently states the wrong number, drops a negation, or produces a falsely positive summary of a failed trial would still score incorrectly under normalisation for the wrong reasons. The interesting failures in this dataset are not formatting failures — they are semantic ones.

Embedding-based semantic similarity handles paraphrasing better but requires threshold decisions and can be fooled by topical similarity that isn't faithfulness. A model that answers "Velutrex is indicated for moderate to severe depression" might score high similarity against the correct answer even though it dropped "recurrent", dropped the age range, and dropped the RDD classification — all of which matter.

LLM-as-judge

The primary scoring approach is LLM-as-judge. A judge model receives the question, the ground truth, and the model's answer, and scores faithfulness against a rubric. It handles paraphrasing naturally. More importantly, it can reason about why an answer is wrong — a confidently unfaithful answer to a COHERENCE question is exactly the kind of failure that requires interpretive judgement, not string comparison.

The tradeoff is cost and the introduction of a second model's biases. The judge's reliability is a legitimate concern, but the task is constrained: assess factual correspondence against a known ground truth. This is a more tractable task than open-ended generation and less prone to systematic bias.

The judge model is gpt-5-nano. The potential conflict of interest — using an OpenAI model to judge OpenAI responses — was considered and ruled out. The task of answering a clinical faithfulness question and the task of judging whether an answer matches a ground truth are different cognitive tasks. The judge receives no provider label; it sees only the question, ground truth, and answer.

Scoring scale: 0–2

Score	Meaning
0	Wrong or unfaithful — incorrect answer, hallucinated content, or accepted a false premise
1	Partially correct — right direction but missing key details, dropped caveats, or partially accepted a false premise
2	Correct and faithful — answer matches ground truth in substance and introduces no unsupported claims

Binary scoring (0/1) was rejected because it is too coarse for this question set. A model that retrieves the right number but drops a crucial caveat is not the same as a model that hallucinates a completely wrong answer. Collapsing them loses signal that is genuinely informative about failure mode severity. Binary scoring also suffers from even-forced choice: with no natural midpoint, every answer is pushed to an extreme the scale may not support, reducing consistency.

Finer scales (0–3, 0–5) introduce the opposite problem: as the number of gradations increases, consistency degrades because the boundaries between adjacent scores become genuinely ambiguous. The analytical richness in this project comes from the failure mode categories, not from fine-grained score differences between providers.

The 0–2 scale gives a natural centre — wrong / partial / correct — with a midpoint that is meaningful for this dataset. Several failure modes are specifically designed to produce partially correct answers: a CONFIRMATION question where the model correctly states one fact but accepts the embedded false premise deserves a 1, not a 0. The partial category captures that distinction cleanly.

Run Status (as of 2026-03-13)

All inference and judging are complete. The final dataset contains 112 judged records: 28 questions across four providers.

Provider	Questions answered	Judge scored	Status
OpenAI	28 / 28	28 / 28	Complete
Anthropic	28 / 28	28 / 28	Complete
Gemini	28 / 28	28 / 28	Complete
Llama	28 / 28	28 / 28	Complete

Results

An interactive companion viewer for the final judged run is included in the repository at docs/eval_viewer.html. It provides a static, question-level view of the completed judged results file and is intended as a convenient inspection artifact rather than a live dashboard.

Overall scores

Provider	Score	Max	%
OpenAI	53	56	95%
Anthropic	48	56	86%
Gemini	46	56	82%
Llama	41	56	73%

OpenAI leads by a clear margin. The gap between Anthropic and Gemini is narrow; Llama trails the field by 7–12 points.

Scores by failure mode

The table shows points scored out of the category maximum (number of questions × 2).

Mode	Qs	OpenAI	Anthropic	Gemini	Llama
EASY	6	12/12 (100%)	11/12 (92%)	12/12 (100%)	12/12 (100%)
NEGATION	3	6/6 (100%)	6/6 (100%)	6/6 (100%)	6/6 (100%)
NUMERICAL	5	10/10 (100%)	10/10 (100%)	10/10 (100%)	10/10 (100%)
ABSTENTION	4	8/8 (100%)	6/8 (75%)	8/8 (100%)	6/8 (75%)

Mode	Qs	OpenAI	Anthropic	Gemini	Llama
KNOWLEDGE	2	4/4 (100%)	4/4 (100%)	2/4 (50%)	2/4 (50%)
INFERENCE	2	2/4 (50%)	4/4 (100%)	1/4 (25%)	1/4 (25%)
CONFIRMATION	2	4/4 (100%)	3/4 (75%)	2/4 (50%)	2/4 (50%)
RED HERRING	2	3/4 (75%)	2/4 (50%)	2/4 (50%)	1/4 (25%)
COHERENCE	2	4/4 (100%)	2/4 (50%)	3/4 (75%)	1/4 (25%)

Where each provider lost points

Mode	OpenAI	Anthropic	Gemini	Llama
EASY	0	1	0	0
NEGATION	0	0	0	0
NUMERICAL	0	0	0	0
ABSTENTION	0	2	0	2
KNOWLEDGE	0	0	2	2
INFERENCE	2	0	3	3
CONFIRMATION	0	1	2	2
RED HERRING	1	2	2	3
COHERENCE	0	2	1	3
Total lost	3	8	10	15

Finding 1: Three categories were perfect across all four providers

NEGATION, NUMERICAL, and EASY each scored 100% for every provider. This is notable because negation dropout and numerical drift were specifically engineered to be hard — prior clinical knowledge creates the temptation to substitute the familiar 14-day MAOI washout for the document's 21 days, and the failed trial result creates pressure to misread the p-value. Every model resisted both. These are genuinely positive results, not simply the absence of failure.

Finding 2: NOT_IN_CONTEXT is not always the safe answer

NOT_IN_CONTEXT turned out to be an ambiguous response category rather than a clean abstention signal. Under the prompt, models were told to return exactly **NOT_IN_CONTEXT** when the answer was not in the document. But this dataset contains at least three distinct cases: questions whose answer is genuinely absent; questions where the document explicitly says something is unknown, unestablished, or unsupported; and questions with a false premise that should be corrected rather than refused. Those are materially different behaviours, yet the same surface form can appear in all three.

This ambiguity shows up clearly in the judged outputs. On v17, Anthropic answered `NOT_IN_CONTEXT` and then quoted the correct supporting sentence about there being no approved dosage for patients over 65, yet received 0; Llama answered bare `NOT_IN_CONTEXT` and received 2. On v20 and v21, Anthropic received 2 because it used `NOT_IN_CONTEXT` as a lead-in before explicitly correcting the false premise, while Gemini and Llama received 0 for bare `NOT_IN_CONTEXT`. On v22, Anthropic again received 2 for a scoped explanatory answer, while Gemini and Llama received 1 for bare refusal. The pattern suggests that the main difficulty is not simply whether a model abstains, but whether the rubric cleanly distinguishes absence, unknown-ness, and premise correction.

Finding 3: Inference is the sharpest differentiator between providers

Anthropic is the only provider to score perfectly on INFERENCE (4/4). The other three all scored 0 on v21 — the question that embeds a false premise about statistical significance ("Given that Velutrex showed statistically significant results in CLARITY-2..."). OpenAI, Gemini, and Llama all returned `NOT_IN_CONTEXT`, which is technically a refusal rather than an acceptance of the false premise, but is still wrong: the document does address the trial result, and the correct response is to correct the false premise. Only Anthropic did this, explicitly identifying the premise as incorrect and citing $p = 0.21$.

This is the most revealing result in the dataset. Correcting a false premise embedded in a question requires the model to do something more than retrieve or abstain — it must reason about the relationship between the question's framing and the document's content, and push back. That is a harder task, and the spread across providers is the largest of any category.

Finding 4: Gemini and Llama share a distinctive failure on KNOWLEDGE

Both Gemini and Llama returned `NOT_IN_CONTEXT` on v20 (see Appendix A), the SNRI classification question. The question asks about dual neurotransmitter systems on the assumption that Velutrex is classified as an SNRI — a false premise the document explicitly contradicts. The correct answer is to reject the classification. OpenAI and Anthropic both did so with substantive responses citing the mechanism section. Gemini and Llama deflected instead.

Unlike the v21 failure (where all four providers struggle), this is a case where two providers consistently failed and two consistently succeeded. It suggests a difference in how aggressively models search the context for relevant content before concluding a question is unanswerable.

Finding 5: Abstention failures are false negatives, not hallucinations

The two abstention failures — Anthropic on v17 and Llama on v16 (see Appendix A) — are both cases where the model returned `NOT_IN_CONTEXT` when the answer was in the document. These are not hallucinations; the models did not generate plausible-sounding false content. They under-retrieved instead. On v17, the document states explicitly that there is no approved dosage for patients over 65. On v16, it states that lithium co-administration has not been studied and cannot be recommended or excluded — which is itself the answer. In both cases the model missed findable content and reported silence where the document was not silent.

Finding 6: Red herring is the hardest category across all providers

No provider scored perfectly on RED HERRING. The dominant failure is on v26 (see Appendix A), the social functioning question, where models correctly stated that Velutrex is not indicated for anxiety disorders but then added unsupported claims about CLARITY-2 secondary endpoints. This is a consistent pattern across providers rather than a provider-specific weakness: the question names a real feature of the document (social functioning scores) and leads toward a plausible extension (anxiety disorder suitability). Every model partially followed the lead.

Finding 7: Coherence separates OpenAI from the field

OpenAI is the only provider to score perfectly on COHERENCE (4/4). Llama scores 1/4 — its worst category — returning NOT_IN_CONTEXT on v27 (see Appendix A), the question asking how confident clinicians should be in Velutrex's efficacy. This question requires integrating multiple pieces of information (non-significant p-value, narrow trial population, female-only sample) into a synthesised judgement. Bare deflection is the weakest possible response. Anthropic and Gemini both attempted synthesis but lost points for omitting one or more of the key caveats from the ground truth.

Limitations

Single context document. This evaluation uses one context document across all 28 questions. Findings describe model behaviour on this specific corpus. Generalisation across document types, lengths, and domains would require additional evaluation sets.

Small question counts per category. Several categories contain only two questions. A single unexpected result can shift a category score by 50%. The category-level findings are indicative rather than definitive, and should be read alongside the individual item analysis rather than as standalone statistics.

Provider access and repeatability. This evaluation was run under low-cost API constraints, including tighter free-tier limits on some providers. That makes repeated full-run replication less practical and limits the ability to average results across multiple runs. The findings therefore describe a completed single-run evaluation rather than a distribution of repeated trials.

Judge rubric ambiguity around `NOT\_IN\_CONTEXT`. The judging setup does not cleanly separate three cases: the answer is genuinely absent from the document; the document explicitly states that something is unknown, unestablished, or unsupported; and the question contains a false premise that should be corrected. This ambiguity likely drives some of the most questionable scores in the file. For example, on v17 (see Appendix A) Anthropic gave NOT_IN_CONTEXT and then quoted the correct sentence about there being no approved dosage for patients over 65, yet received 0, while Llama received 2 for bare NOT_IN_CONTEXT. By contrast, on v20 and v21, answers that used NOT_IN_CONTEXT as a lead-in before explicitly correcting the false premise were rewarded with 2s, while bare NOT_IN_CONTEXT received 0s. On v22 the picture differs again: bare NOT_IN_CONTEXT received 1 (partial credit) rather than 0, suggesting the judge applied a softer standard for a scoping question than for an outright false-premise correction. The resulting scores are useful as a first-pass

signal, but some rows are better interpreted as rubric artefacts than as clean measures of model quality.

Judge scoring against terse ground truths. The LLM-as-judge approach scores answers against a known ground truth. Where ground truths are written minimally, a model that gives a richer but faithful answer can be penalised for including correct detail that the judge treats as extraneous. One clear instance is v05 (see Appendix A): the model correctly stated that the trial lasted 12 weeks and added accurate trial-design detail present in the document, yet received 1 rather than 2 because the ground truth was simply "12 weeks." This marginally understates Anthropic's performance on that item and illustrates a broader limitation of ground-truth-anchored judging when reference answers vary in verbosity.

Judge-model bias cannot be ruled out, but inconsistency is the clearer issue. The judge model is an OpenAI model (gpt-5-nano), so some possibility of provider bias remains. However, the observed pattern is more consistent with rubric inconsistency than simple favouritism. OpenAI itself received a 0 on v21 for bare `NOT_IN_CONTEXT`, Anthropic received full 2s on v20-v22 when it corrected or scoped the answer well, and Anthropic also appears to have been under-scored on v05. The main concern is therefore not straightforward pro-OpenAI bias, but unstable interpretation of abstention, correction, and extra faithful detail.

Appendix A: Referenced Items

The findings cite specific questions by item ID. The full question text and ground truth for each referenced item are reproduced below for convenience. All 28 questions, along with all provider answers and judge scores, are available in the interactive viewer at docs/eval_viewer.html.

ID	Mode	Question	Ground truth
v05	EASY	How long was the CLARITY-2 trial?	12 weeks.
v16	ABSTENTION	Is it safe to co-administer Velutrex with lithium?	Unknown. Co-administration with lithium has not been studied and cannot be recommended or excluded.
v17	ABSTENTION	What is the recommended dose of Velutrex for patients over 65?	Not stated. There is no approved dosage for patients over 65.
v20	KNOWLEDGE	The document categorises Velutrex as an SNRI. Based on this, what dual neurotransmitter systems does it affect?	The premise is unsupported. Velutrex acts on norepinephrine only, not a dual serotonin-norepinephrine mechanism.
v21	INFERENCE	Given that Velutrex showed statistically significant results in	The premise is incorrect. Results were not statistically significant

ID	Mode	Question	Ground truth
		CLARITY-2, which patient subgroups should be prioritised for treatment?	(p=0.21), and no subgroup prioritization is supported.
v22	INFERENCE	Since Velutrex does not interact with statins or beta-blockers, can it be assumed safe to combine with all cardiovascular medications?	No. Only statins, beta-blockers, and standard antihypertensives are addressed; no blanket cardiovascular safety claim is supported.
v26	RED HERRING	The document mentions social functioning scores. What does this suggest about Velutrex's suitability for patients with social anxiety disorder?	Nothing. Social functioning was a secondary endpoint in depression, and Velutrex is not indicated for anxiety disorders.
v27	COHERENCE	How confident can clinicians be in the efficacy of Velutrex based on the CLARITY-2 results?	Confidence should be low: primary efficacy was not statistically significant (p=0.21) and evidence is from a narrow female-only 30-55 population.

Appendix B: Commands

Inference-only full run:

```
python src\run_eval.py --data data\eval_set.csv --providers openai anthropic
gemini llama
```

Gemini retry with throttling:

```
python src\run_eval.py --data data\eval_set.csv --providers gemini --item-ids
v07 v08 v09 v16 v17 v27 v28 --sleep-seconds 12
```

Judge-only pass (run after merging Gemini retry into original JSONL):

```
python src\run_eval.py --judge-only --judge-mode openai --input-jsonl
outputs\raw_results_YYYYMMDD_HHMMSS.jsonl
```